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## SOFTWARE TECHNOLOGIES

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### **Operating Systems:**

Linux  
Ubuntu  
Debian  
Fedora  
CentOS  
Unix  
OpenWrt  
QNX  
Windows  
iOS  
OS X  
Android

### **Containerization & Orchestration:**

Docker  
Docker Compose  
Dockerfile  
Kubernetes  
K3s  
kubectl  
Helm  
minikube  
kubeconfig

### **Cloud Providers & Services:**

Amazon Web Services (AWS)  
Google Cloud Platform (GCP)  
Microsoft Azure  
Google GKE  
Amazon EKS  
Azure AKS  
AWS Bedrock  
AWS SageMaker  
AWS Lambda  
AWS CDK  
Amazon S3  
Amazon RDS  
Amazon DynamoDB  
Amazon CloudFront  
Amazon Kinesis  
Amazon CloudFormation  
Google Cloud Run

Google Pub/Sub  
Google ADK  
Azure AI ML  
Azure Search

### **Infrastructure as Code & Configuration Management:**

Terraform  
Ansible  
YAML

### **Virtualization:**

KVM  
QEMU  
libvirt  
OpenStack  
CloudStack

### **Monitoring & Observability:**

Grafana  
Prometheus  
Jaeger  
OpenTelemetry  
Langfuse  
Opik  
LangSmith  
WandB  
TensorBoard  
Sentry

### **Search Technologies:**

Lucene  
Solr

### **Web Servers:**

Nginx  
Apache HTTP

### **Security & Authentication:**

JWT  
OAuth  
SSL  
TLS  
CORS  
iptables  
ufw

### **Networking & Protocols:**

HTTP  
MQTT  
NATS  
OpenAPI  
Socket.IO  
TCP/IP  
IPv6  
RS232  
Bluetooth Mesh

### **Programming Languages:**

Python  
JavaScript  
TypeScript  
EcmaScript  
Java  
Kotlin  
C  
C++  
Rust  
Go (Golang)  
C#  
Swift  
Objective-C  
VBScript  
R  
Matlab  
Scala  
Groovy  
Dart  
MicroPython

### **Web Frameworks (Node.js/JavaScript):**

Node.js  
Express  
Next.js  
Koa  
Vite

### **Frontend Frameworks & Libraries:**

React  
Redux  
JSX  
Angular  
Svelte  
Vue  
jQuery  
React Native  
Flutter

## **UI/CSS Frameworks:**

HTML  
CSS  
Bootstrap  
Tailwind  
Material UI

## **HTTP Clients & APIs:**

Axios  
Fetch  
HTTPX  
Requests  
GraphQL  
gRPC  
Web Sockets  
RESTful APIs  
SOAP

## **Data Formats:**

JSON  
XML  
BSON  
YAML  
openpyxl  
xlrd

## **Package Managers & Build Tools:**

npm  
yarn  
pip  
webpack  
Babel  
Maven  
Gradle

## **Testing Frameworks:**

Jest  
Jasmine  
Cypress  
Selenium  
Robot Framework  
JUnit  
PyTest  
pytest-cov  
pytest-mock

## **Code Quality & Formatting:**

Black  
Flake8  
PyYAML

### **Go (Golang) Frameworks:**

GORM  
GIN  
Rocket

### **Python Web Frameworks & Libraries:**

Flask  
Flask-CORS  
FastAPI  
Werkzeug  
Gunicorn  
Uvicorn  
Streamlit

### **Python Data & Scientific Libraries:**

NumPy  
Pandas  
Matplotlib  
Plotly  
Seaborn  
SciPy  
Pydantic  
python-dotenv  
tqdm  
aiofiles  
soundfile  
pydub

### **Python AI/ML Libraries:**

sentence-transformers  
datasets  
accelerate  
PEFT  
TRL  
bitsandbytes  
tokenizers  
evaluate  
tiktoken  
sentencepiece  
librosa

### **Java Technologies:**

JDK  
JVM

Struts  
Swing  
JSP  
JNI  
NDK  
ADK

### **Java Frameworks:**

Spring  
Spring Boot  
JDBC  
JPA

### **Application Servers:**

Tomcat  
Jetty  
Undertow  
WildFly

### **Databases (SQL):**

SQL  
PostgreSQL  
pgvector  
MySQL  
Oracle  
Timescale  
Microsoft SQL Server  
SQLite  
H2  
CRUD  
sequelize  
go-pg  
SQLAlchemy  
psycopg2

### **Message Brokers:**

Mosquitto  
RabbitMQ  
Apache ActiveMQ  
Kafka

### **Big Data & Streaming:**

Spark  
PySpark  
Hadoop  
HDFS

### **NoSQL Databases:**

Redis  
MongoDB  
Neo4j  
Cassandra  
HBase  
Mongoose  
PyMongo

### **Machine Learning Frameworks:**

Keras  
TensorFlow  
PyTorch  
TorchScript  
torchvision  
torchaudio  
scikit-learn  
ONNX  
MLLib  
Hugging Face Transformers  
Hugging Face Hub  
AutoML  
Unsloth  
XGBoost  
Gymnasium

### **Neural Networks:**

RNN  
LSTM  
CNN  
NN  
LLM

### **Vector Databases:**

Qdrant  
Chroma  
ChromaDB  
Faiss  
Pinecone  
Weaviate

### **LLM Frameworks & Tools:**

LangChain  
LangChain-OpenAI  
LangChain-Anthropic  
LangChain-Community  
LangChain-Experimental  
LangChain-Ollama  
LangGraph



LlamaIndex  
LiteLLM

### **AI Agent Frameworks:**

CrewAI  
AutoGen  
Agno  
Embabel  
Haystack  
OpenClaw  
Semantic Kernel  
CAMEL-AI  
OWL  
Browser Use  
Strands Agents  
Spring AI  
agent-framework-claude  
DeepAgents

### **AI Development & Coding Tools:**

GitHub Copilot  
Microsoft Copilot  
Langflow  
n8n  
Model Context Protocol (MCP)  
Agent Development Kit (ADK)  
Agent2Agent

### **LLM Providers & Tools:**

OpenAI  
Anthropic Claude  
Ollama  
vLLM  
MLflow

### **Fine-tuning & Optimization:**

LoRA  
QLoRA  
DeepSpeed  
Flash Attention

### **Security & Guardrails:**

Guardrails AI  
LLM Guard  
Pydantic AI

### **Multi-modal AI Libraries:**

Pillow  
OpenCV  
PyPDF2  
PyPDF  
PyMuPDF

### **C/C++ Technologies:**

Qt  
QML

### **Compilers & Debuggers:**

Clang  
LLVM  
GCC  
g++  
GDB  
Yocto

### **GPU Computing:**

CUDA

### **.NET Technologies:**

ASP.NET  
WCF

### **Mobile Development:**

Firebase

### **Data Analysis & Visualization:**

Excel  
JASP  
Looker

### **Version Control & CI/CD:**

Git  
GitHub Actions  
CVS  
SVN  
Jenkins  
Postman  
cURL

### **Data Science Tools:**

Jupyter  
Jupyter Notebook

IPython  
ipykernel  
ipywidgets

### **Workflow Orchestration:**

Apache Airflow  
Dagster  
Kubeflow

### **IDEs & Editors:**

Eclipse  
VS Code  
IntelliJ  
Emacs  
Xcode

### **Shell & Scripting:**

bash scripts